

Lab 2: Decision Tree Modeling and Improvement

1 General Requirements

- Complete the assignment according to the course guidelines.
- Each group must consist of **3 to 5 students**.
- Each group must appoint **one representative** to submit the assignment.
- The submission must be packaged in a single ZIP file named **[GroupID].zip**, including:
 - **Written report:** **[GroupID - Report].pdf**
 - **Source code:** **[GroupID - Code].zip** or a folder containing all code files
 - **Presentation video:** **[GroupID - Video].mp4** or **[GroupID - Video].txt** containing the video link
 - **Optional materials:** **[GroupID - Materials].txt**

2 Objectives

The objectives of this lab are as follows:

- Understand how a decision tree model works for classification or prediction tasks.
- Apply a decision tree algorithm to a real dataset.
- Evaluate the performance of the model using appropriate metrics.
- Analyze the generated tree and interpret its decision rules.
- Propose and implement **2–3 alternative ways** to improve the model performance.
- Practice technical writing, coding, experimentation, presentation, and teamwork skills.

3 Project Requirements

3.1 Project Topic

Each group will work on the topic of **Decision Tree Learning**.

The group must:

- Select **one dataset** for a prediction or classification problem.
- The dataset may be taken from reliable public sources such as:
 - UCI Machine Learning Repository
 - Kaggle Datasets
 - Government open-data portals: European Data Portal, U.S. Government Open Data, WHO Data
 - Other academic or official public datasets: OpenML, Harvard Dataverse
- Build a **decision tree model** on that dataset.
- Write code to train the model and evaluate its performance.
- Compute the model accuracy and error rate.
- Propose and test **2 to 3 different ways** to build or improve the tree in order to obtain better performance.

Note: The focus of this project is not only to run a ready-made model, but also to explain clearly how the tree is constructed, how it behaves on the dataset, and why certain improvements help increase the model performance.

3.2 Required Tasks

Each group must complete the following tasks:

a. Dataset Selection and Description

- Choose one dataset suitable for a decision tree task.
- Describe the dataset, including:
 - Data source
 - Number of samples
 - Number of features
 - Target variable / class labels
- Explain why the dataset is appropriate for decision tree modeling.

b. Baseline Decision Tree Model

- Write code to preprocess the dataset if necessary.
- Split the data into training and testing sets.
- Train a baseline decision tree model.
- Visualize or present the resulting decision tree.
- Compute the model performance using suitable evaluation metrics, depending on the problem type:
 - **For regression:** MAE, MSE, RMSE, and other appropriate metrics.
 - **For classification:** Confusion Matrix, Accuracy, Precision, Recall, F1-score, ROC-AUC, ...

c. Analysis of the Resulting Tree

- Present the generated tree.
- Comment on the structure of the tree.
- Explain important splits or decision rules.
- Provide observations about whether the tree is too simple, too deep, or likely to overfit.

d. Model Improvement

- Propose and implement **2–3 different methods** to improve the model.
- Possible approaches may include:
 - Changing tree depth (`max_depth`)
 - Changing the splitting criterion (e.g., Gini vs. Entropy)
 - Adjusting minimum samples for split or leaf nodes
 - Pruning the tree
 - Feature selection or feature engineering
 - Handling class imbalance
- For each improvement method:
 - Describe the method
 - Present the updated result
 - Compute accuracy and error rate
 - Explain why the method improves or does not improve the model

e. Comparison and Discussion

- Compare the baseline model and the improved models.
- Discuss which method gives the best result.
- Explain the reasons for the improvement in performance.

3.3 Submission Requirements

The submission must include:

- **Code**
 - Full source code used to load data, preprocess data, train the models, generate the tree, and evaluate performance.
 - The code should be clearly organized and commented.
- **Written Report**
 - A report in PDF format describing the dataset, model construction, experimental results, and analysis.
- **Presentation Video**
 - A detailed presentation video explaining the project.
 - The video should clearly present:
 - * the selected dataset,
 - * how the decision tree model is built,
 - * the resulting decision tree,
 - * model accuracy and error rate,
 - * each proposed improvement method,
 - * and the explanation of why the improvements help increase model performance.
 - The video may be submitted as an **.mp4** file or as a text file containing a shared link.

3.4 Written Report Structure

The written report must include the following sections:

- a. **Group Introduction**
 - Group name, student IDs, and member list.
 - Specific contribution of each member.
- b. **Introduction**
 - Brief introduction to decision trees.
 - Objective of the project.
- c. **Dataset Description**
 - Source and description of the dataset.

- Number of samples, features, and target classes.
- Any preprocessing steps performed.

d. **Baseline Model**

- Description of the initial decision tree model.
- Training and testing procedure.
- Resulting tree figure or visualization.
- Accuracy and error rate of the baseline model.

e. **Analysis of the Tree**

- Present the generated decision tree.
- Comment on its structure and key decision nodes.
- Discuss strengths and weaknesses of the baseline tree.

f. **Improvement Methods**

- Present each proposed improvement method separately.
- For each method, include:
 - Description of the method
 - Modified tree or model setting
 - Accuracy and error rate
 - Explanation of why the method improves the result

g. **Comparison of Results**

- Compare the baseline model with the improved models.
- Use tables or figures where appropriate.
- Identify the best-performing method.

h. **Conclusion**

- Summarize what the group learned.
- Present key findings from the experiments.
- Comment on the effectiveness of decision trees for the selected dataset.

i. **References**

- List all references clearly.
- Include dataset source, documentation, textbooks, articles, or libraries used.

4 Important Notes

- The project must include both **code** and **report**.
- A detailed presentation video must also be submitted together with the report and source code.
- The report must contain the **resulting tree**, **comments on the tree**, **each improvement method**, **error rate**, and a clear explanation of **why the improvement increases accuracy**.
- Do not only show final numbers; interpretation and explanation are required.
- The code must be runnable and clearly organized.
- Visual aids such as diagrams, tree figures, tables, and result comparisons are strongly encouraged.

5 Contact

If you have any questions regarding the project, please contact: vntan.work@gmail.com

— GOOD LUCK! —